

Integrating Rational Functions with Partial Fractions

Answer Key

AP Calculus AB/BC

Quick Answers

- | | |
|--|---|
| 1. $\frac{1}{4(x-1)} - \frac{1}{4(x+3)}$ | 5. $\frac{\ln(p)}{10} - \frac{\ln(10-p)}{10}$ |
| 2. $\ln(x-2) - \ln(x+3)$ | 6. $\ln(x) + \operatorname{atan}\left(\frac{x}{2}\right)$ |
| 3. (a) exact mass = $2 \ln(2)$ g, (b) mass to three decimals = 1.386 | 7. (a) exact mass = $4 \ln(2)$ kg, (b) mass to three decimals = 2.773 |
| 4. $\ln(x+2) - \frac{2}{x+2}$ | 8. (a) value on $[1, 3] = \ln\left(\frac{9}{5}\right)$, (b) limit = $\ln(3)$, — |
-

What is verified

11 of 12 answers machine-checked against SymPy, across 8 problems.

— marks an answer only you can judge — problem 8. The worked solution states what a correct response must contain.

Curriculum

Level: AP Calculus AB/BC

FUN-6.D – problems 1, 2, 3, 4, 5, 6, 7

FUN-6.E – problem 8

Difficulty 1–5 of 5 across 12 tagged checks.

1. Decompose $\frac{1}{(x-1)(x+3)}$.

Solution. Write $\frac{1}{(x-1)(x+3)} = \frac{A}{x-1} + \frac{B}{x+3}$, so $1 = A(x+3) + B(x-1)$. Setting $x = 1$ gives $4A = 1$; setting $x = -3$ gives $-4B = 1$.

$$\boxed{\frac{1}{4(x-1)} - \frac{1}{4(x+3)}}$$

2. $\int \frac{5}{(x-2)(x+3)} dx$.

Solution. $5 = A(x+3) + B(x-2)$ gives $A = 1$ at $x = 2$ and $B = -1$ at $x = -3$, so the integrand is $\frac{1}{x-2} - \frac{1}{x+3}$.

$$\boxed{\ln|x-2| - \ln|x+3| + C}$$

Equivalently $\ln \left| \frac{x-2}{x+3} \right| + C$.

- 3.** $r(x) = \frac{6}{(x+1)(x+4)}$ g/min. (a) Exact mass in the first 2 minutes. (b) To three decimals.

Solution (a). $6 = A(x+4) + B(x+1)$ gives $A = 2$ and $B = -2$, so

$$\int_0^2 \left(\frac{2}{x+1} - \frac{2}{x+4} \right) dx = \left[2 \ln \frac{x+1}{x+4} \right]_0^2 = 2 \ln \frac{3}{6} - 2 \ln \frac{1}{4}.$$

$$\boxed{2 \ln 2 \text{ g}}$$

Solution (b).

$$\boxed{1.386}$$

- 4.** $\int \frac{x+4}{(x+2)^2} dx$.

Solution. A repeated factor needs both powers: $\frac{x+4}{(x+2)^2} = \frac{A}{x+2} + \frac{B}{(x+2)^2}$, so $x+4 = A(x+2) + B$, giving $A = 1$ and $B = 2$.

$$\int \left(\frac{1}{x+2} + \frac{2}{(x+2)^2} \right) dx$$

$$\boxed{\ln |x+2| - \frac{2}{x+2} + C}$$

- 5.** $\int \frac{1}{P(10-P)} dP$.

Solution. $1 = A(10-P) + BP$ gives $A = B = \frac{1}{10}$, so

$$\int \frac{1}{P(10-P)} dP = \frac{1}{10} \int \left(\frac{1}{P} + \frac{1}{10-P} \right) dP.$$

$$\boxed{\frac{\ln |P|}{10} - \frac{\ln |10-P|}{10} + C}$$

This is the step that turns the logistic differential equation into something solvable for P .

- 6.** $\int \frac{x^2+2x+4}{x(x^2+4)} dx$.

Solution. The quadratic is irreducible, so its numerator is linear:

$$\frac{x^2+2x+4}{x(x^2+4)} = \frac{A}{x} + \frac{Bx+C}{x^2+4}.$$

Clearing denominators, $x^2 + 2x + 4 = A(x^2 + 4) + (Bx + C)x$. At $x = 0$, $4A = 4$, so $A = 1$; matching x^2 terms gives $B = 0$ and matching x terms gives $C = 2$.

$$\int \left(\frac{1}{x} + \frac{2}{x^2 + 4} \right) dx$$

$$\boxed{\ln |x| + \arctan \frac{x}{2} + C}$$

- 7.** $r(t) = \frac{8}{(t+1)(t+3)}$ kg/yr. (a) Exact mass in 3 years. (b) To three decimals.

Solution (a). $8 = A(t+3) + B(t+1)$ gives $A = 4$ and $B = -4$:

$$\int_0^3 \left(\frac{4}{t+1} - \frac{4}{t+3} \right) dt = \left[4 \ln \frac{t+1}{t+3} \right]_0^3 = 4 \ln \frac{4}{6} - 4 \ln \frac{1}{3}.$$

$$\boxed{4 \ln 2 \text{ kg}}$$

Solution (b).

$$\boxed{2.773}$$

- 8.** $r(x) = \frac{2}{x(x+2)}$ kg/yr. (a) Mass from $x = 1$ to 3. (b) The limit. (c) What it means.

Solution (a). $2 = A(x+2) + Bx$ gives $A = 1$, $B = -1$, so

$$\int_1^3 \left(\frac{1}{x} - \frac{1}{x+2} \right) dx = \left[\ln \frac{x}{x+2} \right]_1^3 = \ln \frac{3}{5} - \ln \frac{1}{3}.$$

$$\boxed{\ln \frac{9}{5}}$$

Solution (b). As $b \rightarrow \infty$, $\frac{b}{b+2} \rightarrow 1$ and its logarithm goes to zero:

$$\boxed{\ln 3}$$

Solution (c) — instructor-judged. The total mass removed over all future time approaches a finite number, $\ln 3$ kilograms, instead of growing without bound: the improper integral converges. So the filter has a lifetime ceiling on what it can ever take out, no matter how long it runs. Full credit needs that finite-total conclusion, not merely the value of the limit.